Project plan

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1 Introduction

Include a very brief synopsis of your literature in the form of a brief background, motivation, and definition of your research problem.

New York City's ongoing transit challenges are marked by inequitable coverage and the overreliance on taxis in underserved areas. Recent literature has applied data-driven techniques—especially taxi GPS data and demographic [1] overlays—to identify so-called "transit deserts," [2] where limited public transportation forces residents to depend on costlier alternatives. These studies emphasize that socio-economic status, distance to transit hubs, and population density are key predictors of taxi use, revealing both the systemic neglect of low-income neighborhoods and the policy imperative to provide equitable transit access.

Innovative solutions have emerged to bridge these mobility gaps without the prohibitive costs of permanent infrastructure expansion. Programs like Dollaride's Clean Transit Access Program (CTAP) use ridership and location data to introduce electric commuter vans into poorly served neighborhoods [3]. Machine learning techniques such as clustering and classification have also guided microtransit pilots and bus network redesigns by identifying latent travel demand. [5] These flexible models allow policymakers to adapt transit routes to meet actual needs, particularly in boroughs like the Bronx where fixed-line infrastructure is politically and economically challenging.

Building on these findings, the proposed research will use large-scale taxi trip datasets [8] to pinpoint where transit coverage fails to meet demand. By analyzing areas with consistently high taxi use and limited access to subways or buses, the study aims to identify prime candidates for expanded bus service or other scalable solutions. Unlike prior efforts focused on long-term infrastructural change, this project emphasizes feasible, near-term interventions—rooted in real behavior patterns—to enhance transportation equity across New York City.

1.1 Sub-Question 1

What benefit would exploration and expansion of public transportation bring to NYC low-income communities?

Hypothesis: We expect that increased public transit access will produce substitution effects, allowing low-income residents to allocate more time and money toward essential goods and services such as food, childcare, and education. Reduced transit costs and commute times will free up resources for other economic and social ventures.

1.2 Sub-Question 2

Which communities will be best suited for expansion?

Hypothesis: We anticipate that communities already identified by New York City as Disadvantaged Community (DAC) zones will be strong candidates for expansion. By combining these official designations with our own taxi pickup and dropoff data, we aim to identify neighborhoods where transit investment would be most impactful.

1.3 Sub-Question 3

Where should expansion occur, and what form should it take?

Hypothesis: Analysis of rideshare and taxi flow data will reveal key corridors of unmet transit demand. Based on observed traffic patterns and prior urban transit studies, we expect that demand-responsive solutions like van-based ridesharing (e.g., Dollaride) will emerge as the most cost-effective method of addressing gaps in underserved areas.

2 Datasets

The Taxi and Limousine commission (TLC) collects data on fares, pickup and dropoff locations, distance traveled, and method used extensively. [8]

The **NYC Disadvantaged Communities (DAC) Census Tract Dataset** will be used in our project and is derived from the 2023 release by the New York State Climate Action Council, accessible through their Disadvantaged Communities (DAC) Criteria portal: [2]. This dataset is vital for us as it will allow us to identify Disadvantaged communities that are in desperate need of affordable transportation, also known as desert transportation zones.

This dataset includes a broad array of variables spanning demographic, housing, health, environmental burden, and energy-related metrics. Each row corresponds to a specific census tract, with features such as total population, unemployment rate, internet access percentage, housing burden (e.g., rent as a percent of income), home energy affordability, percentage of mobile homes, and more. Crucially, the dataset includes a binary DAC designation, indicating whether a tract meets the criteria set forth for being a disadvantaged community. These designations are based on a complex aggregation of indicators intended to highlight cumulative burdens and vulnerabilities. [1]

For this project, the dataset was cleaned to remove corrupted or improperly formatted entries, particularly those in which latitude and longitude values were erroneously inserted in place of textual information (e.g., County or City names). The focus of the analysis is on New York City's five boroughs—Bronx, Brooklyn, Manhattan, Queens, and Staten Island—allowing for a deeper spatial and demographic investigation into how DAC designation correlates with variables of inequality and opportunity across the city.

After this we then plan to take relevant taxi/public transportation datasets and try to identify key areas where new affordable public transportation can be placed.

3 Planned EDA

Briefly describe your planned EDA and general methodological approach.

Firstly, we plan to explore the benefits additional public transportation would allocate to the people of NYC.

Another key component that we wanted to be able to identify were areas that are considered DAC, disadvantaged communities, versus Non-DAC areas. To do this we will perform EDA on the NYC Disadvantaged Communities (DAC) Census Tract Dataset [2]. There are a lot of great columns that we can use, on top of which areas have been identified as DAC and Non-DAC areas. We plan to use key variables to make Bar Charts, Boxplots, and Histograms. On top of this we also plan to create several map plots where key variables are overlaid and then sectioned off into DAC and Non-DAC areas. Some key features that we will be doing this with are Total Population, No College Degree, Unemployment Rate, and potentially others as needed, or that are of key relevance to our project. A feature, yet to be explored, worth delving into would be creating an overlay map using the Households Disabled variable, as these areas might need different forms of transportation provided.

We wanted to look into the areas that taxis made trips. [8] Notably, through our literature review, it was obvious that Manhattan was a heavily trafficked zone with regards to taxis. Reasons for this were concluded with Manhattan, especially lower Manhattan, having the overwhelming majority of NYC's jobs. However, though NYC is densely populated, we wanted to know why these transit deserts existed in the city's suburbs. Part of this analysis will come from measuring the distance of average taxi rides. Motivation for this stems from understanding that Manhattan being a heavily trafficked zone, so if we know taxis are going to Manhattan, do they have a hard time leaving the area because trips are generally too short to reach other regions of the city. The next goal is to get a better understanding of where the pickup and drop off locations are for each ride, and how far the average trip takes for those locations, respectively. To supplement this insight, the next step will involve measuring the number of trips for each location's pickup/drop off code. This will hopefully give us better insight for why taxis are not reaching the other areas.[8] Each location's ID code will then be cross referenced with taxi zone geo-location to give a name to each code. Vectors will be created with each code representing a suburb of NYC. Since the data is represented by a numeric code, cleaning this code and assigning each value a region (Bronx, Brooklyn, etc.) will be imperative to display the number of trips taxis took in each region, regardless if it was for a pickup or a drop-off. [8] Next, we will run a summary comparing the data to each other. This summary will include the average distance of trips for each region, average cost for each trip in a region, most common day that taxis travel to each region, as well as what hour of the day is most common for trips in each region. This will help us better understand taxi routes for NYC. We also plan to create a heatmap of the data regarding pickups for hour of the day and day of the week. This will, theoretically, allow us to draw conclusions for why taxis rides occur more frequently at particular times than others, and the incentive for taxi drivers to have pickups in certain areas over others.[5]

References

[1] Gonzales, E. J., Yang, C., Morgul, E. F., Ozbay, K. (2014). Modeling taxi demand with GPS data from taxis and transit (Report No. CA-MNTRC-14-1141). Mineta National Transit Research Consortium. Available: `https://transweb.sjsu.edu/mntrc/research/Modeling-Taxi-Demand-GPS-Data-Taxis-and-Transit`

[2] New York State Climate Action Council, "Disadvantaged Communities Criteria," 2024. Available: `https://climate.ny.gov/resources/disadvantaged-communities-criteria/`

[3] Dollaride, "Clean Transit Access Program (CTAP)," 2024. Available: `https://www.dollaride.com/ctap`

[4] Safikhani, A., Kamga, C., Mudigonda, S., Faghih, S. S., Moghimi, B. (2017). Spatio-temporal modeling of yellow taxi demands in New York City using generalized STAR models. arXiv preprint arXiv:1711.10090. Available: `https://arxiv.org/abs/1711.10090`

[5] Moosavi, S., Samavatian, M. H., Nandi, A., Parthasarathy, S., Ramnath, R. (2019). Short and long-term pattern discovery over large-scale geo-spatiotemporal data. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery Data Mining (pp. 2905–2913). Association for Computing Machinery. Available: `https://doi.org/10.1145/3292500.3330755`

[6] Yazici, M. A., Kamga, C. N., Singhal, A. (2013). Weather's impact on travel time and travel time variability in New York City. Transportation Research Board 92nd Annual Meeting. Available: `https://www.researchgate.net/publication/255982535_Weather's_Impact_on_Travel_Time_and_Travel_Time_Variability_in_New_York_City`

[7] Diego Armando Hernandez Hernandez A. (2022). Transit Deserts in The Bronx: MTA RPA accessibility study Available: `https://storymaps.arcgis.com/stories/338010c5ce1b4490b163b88a7e04eba6`

[8] New York City Taxi and Limousine Commission. *TLC Trip Record Data*. 2025. `https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page`. Accessed April 13, 2025.